

Gourav Gulia

ML Engineer • MLOps • Generative AI

🏠 Haryana, 131001 | ✉ Email | 🔗 LinkedIn | 🐙 Github | 🌐 Portfolio

Skills

Languages	Python
Machine Learning	Scikit-Learn, TensorFlow, XGBoost, Model Evaluation, Feature Engineering
Generative AI	LangGraph, RAG, Prompt Engineering, Langchain
Computer Vision	OpenCV, FaceNet512, RetinaFace, Image Quality Assessment (PSNR, SSIM, LBP)
MLOps	Apache Airflow, MLflow, DVC, Docker, FastAPI, CI/CD
Software Eng.	REST APIs, Model Serving, Dependency Injection, Factory Pattern, Python Packaging, Modular Architecture
Data & Libraries	pandas, Pandas, NumPy, Pydantic
Databases	SQL, MongoDB, VectorDB (Milvus, FAISS, Weaviate)
Tools	Git, Linux, Streamlit

Experience

EY (Ernst & Young)	Jun 2025
Sr. Analyst	Gurugram
- Identified malpractice patterns and compliance gaps across large-scale government examination datasets using pandas, statistical analysis, and forensic validation techniques on multimodal data including facial images, demographic records, and system logs. - Developed and deployed production-grade face verification, image quality analysis, and morphing detection pipelines for government clients including SSC, HSSC, and NHA, leveraging FaceNet512, RetinaFace, PSNR, SSIM, and LBP to improve verification accuracy across challenging image conditions. - Designed a weighted ensemble inference framework optimized through A/B Testing, LCB/UCB strategies, and ROC-based threshold tuning, reducing false positives by nearly 11% while improving verification consistency. - Built an in-house Text Similarity & Impersonation Detection System using Jaro-Winkler similarity, phonetic matching, and TF-IDF vectorization, improving duplicate identity detection efficiency by approximately 25% during forensic investigations. - Re-architected legacy ML solutions into a modular, reusable, and API-driven framework using FastAPI, while implementing workflow orchestration with Apache Airflow and experiment tracking through MLflow, significantly improving deployment scalability and reproducibility. - Optimized production inference pipelines through multiprocessing, hyperparameter tuning, and memory-efficient processing techniques including dtype downcasting, reducing inference latency by approximately 35% and memory utilization by 22%. - Collaborated with cross-functional teams and government stakeholders to deliver enterprise forensic AI solutions aligned with regulatory compliance, risk governance, and standardized investigation workflows across large-scale public sector projects.	
Netmax	Oct 2024 - Feb 2025
Jr. Data Scientist	Chandigarh
Built a modular data preprocessing pipeline leveraging MLflow, DVC, and Pandas, which reduced data drift and noise by 20%, boosting downstream model accuracy across production workflows.	

Education

M.Sc. Data Science	Jul 2022 – May 2024
Chandigarh University	
B.Sc. Applied Science	Jul 2019 – May 2022
Delhi University	

Featured Projects

Aurelius - Autonomous Multi-Agent Research & LLMOps System | Links •

Jun 2026 - Aug 2026

- Architected a stateful **multi-agent research workflow** using LangGraph and LangChain with 5 specialized personas: Lead, Researcher, Synthesizer, Writer, and Reviewer, enabling autonomous query decomposition, cross-source evidence gathering, and self-correcting revision loops for citation-backed technical reports.
- Engineered an enterprise-grade **LLMOps observability and evaluation framework** with LangSmith distributed tracing, granular token and latency tracking for Groq LPU inference, and automated quality scorecards measuring citation density, factual consistency, and evidence grounding with a strict zero-link hallucination guarantee.
- Built a production-ready **Python 3.12 architecture** using Pydantic v2 validation, provider-abstracted LLM factories, resilient search and scraping fallback pipelines, and structured state management for reliable multi-agent execution.
- Developed a real-time **Streamlit reasoning dashboard** and production containerization workflow using multi-stage Docker and UV, backed by a **28-test automated test suite** covering core workflow, validation, and reliability components.

End-to-End AI-Powered Burnout Risk Prediction | Links • • • • •

Jun 2026 - Aug 2026

- Engineered an end-to-end **student burnout prediction system** covering data validation, feature engineering, model training, evaluation, and production inference.
- Built a configurable **ML experimentation framework** for systematic model comparison and hyperparameter optimization across multiple classification algorithms.
- Implemented a **reproducible, artifact-driven ML workflow** using DVC and MLflow for versioned datasets, experiment tracking, model management, and pipeline reproducibility.
- Productionized the model with **FastAPI, Streamlit, and Docker**, deploying a live prediction API and interactive UI with a publicly available container image.

CIP: Candidate Intelligence Platform

Jan 2026 - Mar 2026

- Developed a **Ensembled candidate risk prediction model** using structured examination metadata and forensic indicators, providing the foundation for downstream explainable AI workflows.
- Architected a modular, configuration-driven **AI risk intelligence platform** at EY GPS Assurance using reusable pipeline components, artifact-based communication, **Dependency Injection**, and YAML-driven configurations, enabling scalable and maintainable ML workflows.
- Built a production-grade **MLOps pipeline** leveraging **Apache Airflow, DVC, MLflow, Docker, and FastAPI** for automated orchestration, experiment tracking, data versioning, reproducible model training, and scalable inference.
- Developed an AI-powered **Investigation Copilot** using **LangGraph**, advanced **prompt engineering**, and a **Retrieval-Augmented Generation (RAG)** architecture to retrieve SOPs, historical investigation cases, and policy documents for explainable, evidence-backed decision support.
- Designed an interactive **Streamlit** investigation dashboard integrating real-time ML predictions with conversational AI, enabling investigators to analyze candidate risk, understand model reasoning, and receive context-aware investigation recommendations through a production-ready interface.

Inspector: In-House Forensic Analytics Library for EY

Jul 2025 – Aug 2025

- Designed an in-house modular **Python package (.whl)** and **Dockerized forensic analytics library** at EY for scalable image and text-based ML workflows, enabling reusable and production-ready forensic pipelines.
- Engineered a high-performance **face retrieval system** using **VectorDB** and **HNSW-based ANN indexing**, enabling near real-time similarity search across **1M+ embeddings** with optimized retrieval latency.
- Built a scalable **text forensic analytics engine** using **Jaro-Winkler similarity, TF-IDF vectorization, and nearest neighbor search** to detect impersonation and duplicate identity patterns across large-scale datasets.
- Accelerated ML pipeline performance using **multiprocessing, memory-efficient processing**, and advanced software engineering principles including **Dependency Injection** and **Factory Pattern**, improving maintainability and execution efficiency.

Research Publications

Liver Disease Prediction Using Ensemble Learning

- Developed a predictive system for **liver disease detection** using multiple **machine learning algorithms**, enabling early-stage diagnosis through data-driven insights.
- Performed **comparative analysis** across models to evaluate performance using metrics such as **accuracy, precision, and recall**, achieving up to **95% accuracy**.
- Applied comprehensive **data preprocessing, feature selection, and model validation** techniques to improve generalization and robustness on medical datasets.
- Demonstrated strong understanding of **healthcare analytics, statistical modeling, and ML evaluation frameworks**, contributing to a research-backed solution for disease prediction.